



Pharm*Achieve*

MENTAL HEALTH DRUG CHARTS

Qualifying Exam Preparatory Course

LEGEND

5-HT Serotonin

ACEi Angiotensin-Converting Enzyme Inhibitor

ARB Angiotensin II Receptor Blocker

ACH Anticholinergic

AEs Adverse Effects

ASA Acetylsalicylic Acid

BID Twice daily

C/I Contraindication

CNS Central Nervous System

CYP Cytochrome P450

ECG Electrocardiogram

EPS Extrapyrarnidal Symptoms

GABA Gamma-Aminobutyric Acid

GI Gastrointestinal

HTN Hypertension

IM Intramuscular

MAOI Monoamine Oxidase Inhibitor

MOA Mechanism Of Action

NE Norepinephrine

NSAID Nonsteroidal Anti-Inflammatory Drug

P-gp P-Glycoprotein

SERT Serotonin Transporter

SNRI Serotonin-Norepinephrine Reuptake Inhibitor

SSRI Selective Serotonin Reuptake Inhibitor

SELECTIVE SEROTONIN REUPTAKE INHIBITORS

Examples	MOA	Side effects	Interactions
Escitalopram, citalopram, fluoxetine, fluvoxamine, paroxetine, sertraline	Bind to SERT, prevent serotonin reuptake	Headache, fatigue, orthostatic hypotension, insomnia, GI effects (e.g. abdominal pain, constipation, decreased or increased appetite, dyspepsia), ACH effects, sexual dysfunction, agitation	• Serotonin syndrome with MAOIs/other serotonergic medications • NSAIDs and antiplatelets: ↑risk of bleeding • SSRIs inhibit some CYP450 enzymes

SEROTONIN NOREPHINEPHRINE REUPTAKE INHIBITOR

Examples	MOA	Side effects	Interactions
Venlafaxine, desvenlafaxine, duloxetine, levomilnacipran	Block reuptake of serotonin and NE	Headache, insomnia, sweating, nausea, dry mouth, sexual dysfunction, sleepiness	• Serotonin syndrome with MAOIs or other serotonergic medications • Dose-related hypertension with venlafaxine at approximate doses > 225-300 mg (do not use if HTN is uncontrolled) • NSAIDs and antiplatelets: ↑risk of bleeding • Duloxetine inhibits CYP2D6

DUAL ACTION ANTIDEPRESSANTS

Examples	MOA	Side effects	Interactions
Bupropion	Inhibits the reuptake of NE and dopamine	• Agitation, anxiety, insomnia, pruritus, nausea, ↑seizures, ↓appetite • BID dosing separate by <u>at least</u> 8 hours • C/I: seizure disorder, anorexia/bulimia nervosa, electrolyte disturbances	Avoid MAOIs, may ↑cyclophosphamide levels
Mirtazapine	Stimulates NE and serotonin release and inhibits 5-HT ₂ and 5-HT ₃ receptors	Weight gain, dry mouth, and sedation, increased serum cholesterol, xerostomia, increased appetite, constipation	Avoid MAOIs, QT prolongation drugs, alcohol/CNS depressants may ↑sedation

SEROTONIN MODULATORS

Examples	MOA	Side effects	Interactions
Trazodone	Potent serotonin receptor antagonist with weak serotonin reuptake inhibitory effects	Drowsiness, sedation, orthostasis, headache, fatigue, blurred vision, nausea, vomiting, xerostomia, constipation, diarrhea (slightly less common compared with constipation) priapism (rare)	• Avoid MAOIs/ CNS depressants • Antihypertensive agents • CYP3A4 inhibitors
Vortioxetine	Serotonin modulation and transporter inhibition	Sexual dysfunction, nausea, vomiting, flatulence dizziness, diarrhea, constipation (slightly less common compared to diarrhea), xerostomia	• Avoid MAOIs (fatal) & serotonergic agents • Bleeding risk with warfarin, ASA and other antiplatelet agents

TRICYCLIC ANTIDEPRESSANTS

Examples	MOA	Side effects	Interactions
Amitriptyline, nortriptyline , clomipramine, desipramine , imipramine, doxepin	Act primarily as serotonin-NE reuptake inhibitors	Anticholinergic effects (e.g. urinary retention, dry mouth, constipation), worsening of heart arrhythmias, heart block, QT prolongation, sedation, weight gain, orthostasis, sexual dysfunction	• Avoid ACH drugs • CNS depressants • MAOIs (fatal) • SSRI/SNRI → serotonin syndromes • Highly toxic in overdose due to cardiac effects



nortriptyline and **desipramine** are secondary amines and as a result have a lower incidence of side effects

MONOAMINE OXIDASE INHIBITORS

Examples	MOA	Side effects	Interactions
• Moclobemide is a reversible MAO_A inhibitor • Phenelzine and tranylcypromine are non-selective irreversible MAO_A and MAO_B inhibitors	Inhibits the enzyme monoamine oxidase	Insomnia, palpitations, tachycardia, orthostatic hypotension, sexual dysfunction (phenelzine and tranylcypromine), ACH effects, weight gain, cardiac effects, headaches, nausea, vomiting	• Avoid tyramine-rich foods with phenelzine and tranylcypromine • Sympathomimetic agents • Levodopa • Other serotonergic drugs

Switching drugs: General rules

- Irreversible MAOI → other antidepressants: 14-day washout period between discontinuation and initiation of new drug (exception: moclobemide washout requires 1 day)
- Other antidepressants → Irreversible/reversible MAOI: 14 days (exceptions include fluoxetine washout which is 5 weeks)
- SwitchRx.com

SECOND GENERATION ANTIPSYCHOTICS: OLANZAPINE

Formulations	MOA	Side effects	Interactions
Regular and orally disintegrating tablets, IM injection	• 5-HT₂ and D_{1,2,3,4} dopamine receptor antagonist	Orthostatic hypotension, drowsiness, insomnia (less often compared to drowsiness), EPS (e.g. akathisia, parkinsonism, akinesia, tremors), weakness, headaches, fatigue, increased appetite, xerostomia, dyspepsia, constipation, ↑serum AST, ↑serum ALT, ↓serum bilirubin, weight gain, hyperglycemia, fever, cough, rhinitis, nasopharyngitis, pharyngitis, anticholinergic effects	• CYP1A2 inhibitors/inducers • CYP2D6 inhibitors/inducers (to a lesser extent) • Fluvoxamine, fluoxetine and ciprofloxacin • Omeprazole • Smoking

SECOND GENERATION ANTIPSYCHOTICS: QUETIAPINE

Formulations	MOA	Side effects	Interactions
Immediate and extended release	Inhibits 5-HT₂ and D₁, D₂ receptors	Sedation, dizziness, agitation, headaches, weight gain, increased appetite, orthostatic hypotension, tachycardia, palpitations, anticholinergic effects, hyperglycemia, dyslipidemia, ↑risk of cataract formation, xerostomia, constipation, decreased hemoglobin, EPS (e.g. tremors), ↑serum alanine and aspartate aminotransferases, ↑serum transaminases, nasal congestion, paranasal sinus congestion, pharyngitis, rhinitis, upper respiratory tract infection, fever	- CNS depressants - CYP3A4 Inhibitors/Inducers

SECOND GENERATION ANTIPSYCHOTICS: RISPERIDONE

Formulations	MOA	Side effects	Interactions
Regular and disintegrating tablets, long-acting injection	Combined antagonist activity at D₂ dopamine receptor and 5-HT₂ serotonin receptor	EPS (e.g. parkinsonism, tremors, akathisia), orthostatic hypotension, hyperprolactinemia, nausea, vomiting, constipation, upper abdominal pain, nasopharyngitis, cough, rhinorrhea, fever, increased appetite, weight gain	- Carbamazepine - Inhibitors/inducers of CYP2D6, CYP3A4 and P-gp

SECOND GENERATION ANTIPSYCHOTICS: PALIPERIDONE

Formulations	MOA	Side effects	Interactions
Extended-release tablets, long-acting injection	Antagonist activity at D₂ 5HT₂ receptors	Tachycardia, drowsiness, EPS (e.g. akathisia, parkinsonian-like syndrome, dystonia, tremors, hyperkinesia), ↑serum prolactin, ↑serum triglycerides, ↑LDL, increased cholesterol, ↓HDL, hyperglycemia, vomiting, weight gain	- Strong CYP3A4 inducers - QTc prolonging agents - P-gp substrate

SECOND GENERATION ANTIPSYCHOTICS: ARIPIPAZOLE

Formulations	MOA	Side effects	Interactions
Tablets, long-acting injection	• Partial agonist at D₂, 5-HT_{1A} receptors • Antagonist at 5-HT_{2A} receptors	↑serum triglycerides, ↑LDL, ↑cholesterol, ↓HDL, EPS (tremor, akathisia, dystonia), headaches, weight gain, orthostatic hypotension, tachycardia, nasal congestion, upper respiratory tract infection, neutropenia, nausea, constipation, increased serum glucose, drowsiness, dizziness	• Strong inducers/inhibitors of CYP2D6/3A4 can ↑/↓ levels • Carbamazepine (↓levels of aripiprazole) • Fluoxetine, erythromycin, quinidine, protease inhibitor and paroxetine (↑aripiprazole)

SECOND GENERATION ANTIPSYCHOTICS: ZIPRASIDONE

Formulations	MOA	Side effects	Interactions
Capsules	• Antagonism at 5-HT_{2A}, > antagonism at D₂ • 5-HT_{1D} antagonist • 5- HT_{1A} agonist	Drowsiness, dizziness, headaches, EPS (e.g. dyskinesias, tremors, akinesia), orthostatic hypotension, hypertension, tachycardia, bradycardia, weight gain, constipation, vomiting, xerostomia, diarrhea, sialorrhea, QTc prolongation	• CYP3A4 (carbamazepine ↓levels of drug) • QT prolonging agents

SECOND GENERATION ANTIPSYCHOTICS: LURASIDONE

Formulations	MOA	Side effects	Interactions
Oral tablets	Antagonist activity at D₂ and 5-HT_{2A} receptors	↑serum triglycerides, ↑cholesterol, ↑serum glucose, nausea, vomiting, dyspepsia, xerostomia, diarrhea, viral infections, urinary tract infection, rhinitis, nasopharyngitis, influenza, agitation, EPS (e.g. akathisia, parkinsonian-like syndrome), ↑serum creatinine, drowsiness, insomnia, tachycardia, orthostatic hypotension, weight gain, ↑serum prolactin	- QT prolonging drugs - C/I: Strong CYP3A4 inhibitors/inducers thus use lower doses with moderate CYP3A4 inhibitors

SECOND GENERATION ANTIPSYCHOTICS: ASENAPINE

Formulations	MOA	Side effects	Interactions
Sublingual tablets (↓bioavailability if swallowed)	Antagonist activity at D₂ and 5-HT_{2A} receptors	Drowsiness, dizziness, insomnia, EPS (e.g. akathisia, drug-induced Parkinsonism), fatigue, headaches, ↑serum triglycerides, ↑cholesterol, ↑serum glucose, ↓HDL cholesterol, oral hypoesthesia, ↑creatinine phosphokinase in blood specimen, tachycardia, hypertension, orthostatic hypotension, ↑appetite, abdominal pain, dysgeusia, constipation, diarrhea (less common than constipation), nasopharyngitis, upper respiratory tract infection, QTc prolongation	- Metabolized via CYP1A2 oxidation and UGT1A4 glucuronidation - Strong CYP1A2 inducers/inhibitors - QT prolonging agents

SECOND GENERATION ANTIPSYCHOTICS: CLOZAPINE

Formulations	MOA	Side effects	Interactions
Tablets	Selectively blocks DA and 5-HT receptors	Tachycardia, hypotension, hypertension, drowsiness, dizziness, sedation, insomnia (occurs less than sedation), vertigo, sialorrhea, weight gain, constipation, nausea, vomiting, dyspepsia, xerostomia, fever, myocarditis, cardiomyopathy, diaphoresis, leukopenia, neutropenia, eosinophilia, agranulocytosis, visual disturbances, EPS (e.g. tremors, akathisia, pseudo-parkinsonism), restlessness, nightmares, seizures, dyslipidemia, hyperglycemia	- CYP1A2 inhibitors - CYP3A4 inhibitors may ↑clozapine levels - CYP1A2 or CYP3A4 inducers (e.g. carbamazepine, phenytoin, rifampin) may ↓clozapine levels (tobacco smoke also CYP1A2 inducer)

ANTIPSYCHOTICS: COMPARATIVE DRUG CHART

Counsel on antipsychotic-associated body temperature dysregulation

Drug	Sedation	Insomnia	EPS	Weight gain	Metabolic abnormalities	Hyperprolactinemia	CV effects
Aripiprazole	+	+++	+++	++	+	+	++
Asenapine	+	+	++	+	+	+	++
Brexipiprazole	+	+	0	++	+	+	+
Clozapine	+++++	0	0	+++++	+++++	0	+++
Lurasidone	++	+	++	+	+	+	++
Olanzapine	++++/ +++	0	+	+++++	+++++	++	++
Paliperidone	+	+++	++++/ +++	+++/ ++	+	++++/ +++	++
Quetiapine	++++	0	0	++++	+++	0	++
Risperidone	++	++	++++	+++	+	++++	++
Ziprasidone	++	++	++	+	+	++	+++

LEGEND: 0: equivalent to placebo ; +: minimal or none ; ++: very low ; +++: low ; ++++: moderate ; +++++: high

FIRST GENERATION ANTIPSYCHOTICS

Examples	MOA	Side effects	Interactions
• Low potency: chlorpromazine, methotrimeprazine • Intermediate potency: loxapine, perphenazine, zuclopenthixol • High potency: flupentixol, fluphenazine, haloperidol, pimozide, trifluoperazine	• Dopamine pathway antagonist	• Lower potency FGAs have more sedation, CV effects, anticholinergic effects, and weight gain • High potency agents have more EPS (e.g. parkinsonism, tardive movement disorders such as tardive dyskinesias), neuroleptic malignant syndrome, and elevated prolactin levels	• Additive effects with CNS depressants, anticholinergics, α-adrenergic antagonists • Inhibitors and inducers of CYP P450 enzymes (e.g. carbamazepine)

● D₂ receptor binding affinity determines potency

MOOD STABILIZERS: LITHIUM

Formulations	MOA	Side effects	Interactions
Oral immediate release tablets and capsules, extended-release tablets, oral solutions	MOA not clearly understood	Hypothyroidism, hyperthyroidism, hyperparathyroidism, hypercalcemia (including secondary to hyperparathyroidism), ECG changes (e.g. arrhythmias), bradycardia, cold extremities, edema, hypotension, myxedema, syncope, alopecia, psoriasis, worsening of existing psoriasis, skin rash, xeroderma, folliculitis, diabetes insipidus, dehydration, glycosuria, hyperglycemia, increased thirst, polydipsia, weight gain, weight loss, leukocytosis, local skin discolouration, tremors, tinnitus, ↓creatinine clearance, glomerulopathy, nephrotic syndrome	• NSAIDs • ACEi/ARBs • Thiazide diuretics

MOOD STABILIZERS: DIVALPROEX/VALPROIC ACID

Formulations	MOA	Side effects	Interactions
Immediate release oral tablets, delayed release oral tablets, extended release (24-hour) tablets, IV injection	Enhanced GABA neuro-transmission	Headaches, drowsiness, dizziness, nervousness, pain, insomnia, alopecia, nausea, vomiting, xerostomia, abdominal pain, diarrhea, dyspepsia, anorexia, weight loss, weight gain (more than weight loss) thrombocytopenia, tremors, asthenia, diplopia, visual disturbances (e.g. blurred vision, amblyopia), flu-like symptoms, viral infection, menstrual irregularities, nystagmus, ↑serum alanine and aspartate aminotransferase, bronchitis, pharyngitis, rhinitis, pneumonia, dyspnea, epistaxis, fever, osteoarthritis, bone marrow suppression, pancreatitis, hepatotoxicity, polycystic ovarian syndrome	↑ Drug levels: • Erythromycin • Clarithromycin ↓ Drug levels: • Phenytoin • Rifampin • Carbapenems • Carbamazepine

MOOD STABILIZERS: LAMOTRIGINE

Formulations	MOA	Side effects	Interactions
Tablets (swallow or chewable)	Sodium channel blocker	Nausea, chest pain, edema, insomnia, fatigue, ataxia, dizziness, irritability, amnesia, depression, migraines, skin rash, increased libido, vomiting, dysmenorrhea, urinary frequency, rectal hemorrhage, myalgia, rhinitis, bronchitis, respiratory tract infection, fever, cough, diplopia, hepatotoxicity (rare), leukopenia (rare)	• Valproic acid • Carbamazepine • Phenytoin • Combined oral contraceptives • Progestin only pill

MOOD STABILIZERS: CARBAMAZEPINE

Formulations	MOA	Side effects	Interactions
Tablet (swallow), chewable, extended-release oral tablet	Voltage gated- sodium channel blocker	Dizziness, drowsiness, ataxia, nausea, vomiting, hypertension, pruritus, skin rash, constipation, weakness, tremors, blurred vision, alopecia, menstrual disturbances, hyponatremia, hepatotoxicity, nystagmus, tinnitus, hirsutism	Many drug interactions as it induces CYP enzymes (e.g. potent inducer of 1A2, 3A4, 2C9, among others)

BENZODIAZEPINES

Formulations	MOA	Side effects	Interactions
Alprazolam, clonazepam, diazepam, lorazepam, midazolam May be available as oral tablets (to swallow), sublingual tablets, injectables	GABA receptor binding	Memory/cognitive impairment, drowsiness (and risk of falls), sedation, muscle weakness, ataxia, tolerance/dependence	• Alcohol • CNS depressants • Avoid in patients with history of drug/alcohol abuse • Caution with CYP inducers and inhibitors

AZAPIRONES: BUSPIRONE

Formulations	MOA	Side effects	Interactions
Oral tablet (to swallow)	Unknown MOA; high affinity for serotonin 5-HT _{1A} and 5-HT ₂ receptors (without affecting benzodiazepine-GABA receptors) and also has a moderate affinity for dopamine D ₂ receptors	Dizziness, chest pain, drowsiness, headache, abnormal dreams, ataxia, excitement, outburst of anger, restlessness, confusion, diaphoresis, skin rash, weakness, tremors, blurred vision, tinnitus, nasal congestion; has NO abuse potential	• Avoid with MAOIs • Buspirone is a substrate of CYP3A4 (major) and CYP2D6 (minor) thus use caution when combining with strong CYP inducers or inhibitors

NON-SELECTIVE BETA- BLOCKERS: PROPRANOLOL

Formulations	MOA	Side effects	Interactions
Oral tablet (to swallow), oral solution, extended-release capsule, intravenous (injectable)	Beta ₁ and beta ₂ receptor antagonist	Cold extremities (higher frequency in infants), angina pectoris, bradycardia, cardiac failure, hypotension, sleep disorder (higher frequency in infants), agitation (higher frequency in infants), fatigue, dizziness, nightmares (higher frequency in infants), contact dermatitis, pruritus, skin rash, dysglycemia (hyperglycemia or hypoglycemia), hyperlipidemia, hyperkalemia, diarrhea (higher frequency in infants), decreased appetite (higher frequency in infants), bronchospasm, bronchitis (higher frequency in infants and is associated with cough, fever, diarrhea, and vomiting)	Propranolol is a substrate of: CYP1A2 (major), CYP2D6 (major), CYP2C19 (minor), CYP3A4 (minor), thus use caution when combining with CYP inducers or inhibitors

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CHANGE LOG

March 2025

- Slide 18: Updated chart based on reference from rxtx

May 2025

- Slide 11: Modified interactions

August 2025

- Slides 9 and 26: Minor edits to interactions

November 2025

- Minor formatting/grammatical changes made

March 2026

- No content changes made